

Andres Torres

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EDUCATION

The University of Texas at Austin

Expected December 2027

B.S. Electrical & Computer Engineering | Engineering Honors Program | Dell Scholar | GPA: 3.72

Austin, TX

Relevant Coursework: ECE 319K Embedded Systems, ECE 312 Software Design I, ECE 422C Software Design II, ECE 313 Linear Systems & Signals, M 427J Differential Equations & Linear Algebra, M 325K Discrete Mathematics

TECHNICAL SKILLS

Languages: C/C++, Python, Java, JavaScript (Node.js), VHDL, Verilog, PHP, Julia | **Fluent:** English, Spanish

CS Fundamentals: Multithreading, Concurrency, Memory Management, Data Structures, Algorithms, TCP/IP Networking, FSM Design

Data & Analytics: Data Pipelines, ETL, Pandas, NumPy, TF-IDF/BM25, Inverted Index, Web Scraping, Automated Reporting

AI / ML: LLM Deployment & Fine-Tuning, Neural Networks (from scratch), NLP, Relevance Scoring, RAG

Tools & Platforms: Git, Linux, VS Code, Docker (familiar), Supabase, Next.js, React, Excel, Vivado

Additional: Willing to relocate anywhere in the U.S.

EXPERIENCE

Unimex Group

Summer 2026

AI Software Engineering Intern

Laredo, TX

- Deployed and fine-tuned a local LLM to serve as a centralized AI system for the company, personalizing it to handle technical workflows across departments including customs, warehouse operations, billing, and logistics
- Developed Python automation scripts for the customs team to segregate FTZ-bound items from general shipments, eliminating manual classification and reducing processing time across high-volume import operations
- Automated purchase order generation workflows in Python, integrating with existing ERP and CRM systems to reduce manual data entry and improve cross-departmental synchronization
- Managed database migrations and schema updates while ensuring data integrity and minimizing downtime.
- Architecting a facial recognition pipeline integrated with timeclock hardware and QuickBooks payroll to autonomously capture attendance data and process payroll entries without manual input

Texas Marine Robotics

Fall 2025 – Present

Lead Electrical Engineer

Austin, TX

- Lead embedded firmware development and electrical design for an autonomous marine vehicle, writing C/C++ for real-time sensing, actuation, and power management subsystems

Web Developer

Summer 2023 – Present

Freelance

Remote

- Deliver custom full-stack web applications and KPI dashboards for multiple clients using Next.js and React, managing full development cycles from requirements through deployment

PROJECTS

Multiplayer Networked Battleship · Java / React

Summer 2025

- Engineered a full-stack multiplayer game over raw WebSockets with a multithreaded Java server using a producer-consumer model — dedicated ClientHandler threads feed a WriterThread draining a LinkedBlockingQueue, fully decoupling game logic from I/O
- Implemented thread-safe state management with Java's synchronized keyword and an explicit phase-state machine (WAITING → PLAYING → DONE) to eliminate race conditions across concurrent events; built a custom JSON serializer handling all 10 protocol message types without external libraries

Memory Allocator (malloc/free) · C

Summer 2025

- Implemented a heap allocator in C from scratch using sbrk(), replacing malloc/free/realloc/calloc with an explicit free list and O(1) boundary-tag coalescing; developed a 23-test suite including a 1 MB stress allocation and heap integrity checker — achieved 23/23 passing

Full-Text Search Engine · Python

Summer 2025

- Built an inverted index search engine with TF-IDF and Okapi BM25 ranking, Porter stemmer, stop-word filtering, boolean query parser (AND/OR/NOT), and phrase search via positional adjacency — indexed 10 documents with 207 unique terms and sub-millisecond query response

ACTIVITIES & ORGANIZATIONS

Society of Hispanic Professional Engineers (SHPE) | IEEE | Project M.A.L.E.S./I.M.P.A.C.T. Mentor | B.E.S.T Robotics